

Suggested teaching flow: Excel Data Analysis – Practical Workshop

Use a simple **Sales & Profit dataset** throughout:

Date	Region	Product	Units	Unit Price	Sales	Cost	Profit	Ad Spend
01-Jan	North	Laptop	10	50,000	500,000	420,000	80,000	20,000
02-Jan	South	Laptop	12	50,000	600,000	500,000	100,000	25,000
03-Jan	West	Tablet	20	25,000	500,000	410,000	90,000	18,000
04-Jan	East	Mobile	30	20,000	600,000	480,000	120,000	30,000
05-Jan	North	Mobile	35	20,000	700,000	550,000	150,000	32,000

For actual practice, I would expand this to **500–1,000 rows**.

1. Data Analysis: start with the business question

Don't start by teaching formulas. Start with:

"Management has given us sales data. What can Excel tell us that will help management make a decision?"

Teach learners to move through:

Raw Data → Clean → Summarize → Analyze → Visualize → Interpret → Decide

Give them questions such as: Which region generates the highest sales? Which product is most profitable? Is advertising related to sales? Are sales increasing over time? What sales level is required to achieve a target profit?

This gives purpose to everything that follows.

2. Descriptive Statistics

Take the **Sales** column and ask:

"Before building any model, what does a typical sales day look like?"

Teach:

=AVERAGE(F2:F501)

=MEDIAN(F2:F501)

=MODE.SNGL(F2:F501)

=MIN(F2:F501)

=MAX(F2:F501)

=STDEV.S(F2:F501)

=VAR.S(F2:F501)

=COUNT(F2:F501)

Then show the faster method:

Data → Data Analysis → Descriptive Statistics

Select Sales as the input range and generate Summary Statistics.

Explain the interpretation, not just calculation:

Mean = typical average sales

Median = middle observation

Standard deviation = how much sales fluctuate

Minimum/Maximum = range of observed performance

A good question for students:

"Two regions both average ₹5 lakh sales. Region A has SD ₹40,000 and Region B has SD ₹2 lakh. Are they really performing similarly?"

This introduces variability.

3. Covariance

Now compare **Advertising Spend and Sales**.

=COVARIANCE.S(I2:I501,F2:F501)

Explain:

Positive covariance: variables generally move together.

Negative covariance: when one increases, the other tends to decrease.

Near zero: little linear co-movement.

The important teaching point is that the magnitude of covariance depends on the units, which makes comparison difficult. That naturally leads to **correlation**.

4. Correlation

Ask:

"Does higher advertising expenditure tend to be associated with higher sales?"

Use:

=CORREL(I2:I501,F2:F501)

Interpret roughly:

Correlation Interpretation

+1 Perfect positive relationship

Correlation Interpretation

+0.8	Strong positive relationship
0	No linear relationship
-0.8	Strong negative relationship
-1	Perfect negative relationship

Then use:

Data → Data Analysis → Correlation

Have students create a correlation matrix for:

Units | Unit Price | Sales | Cost | Profit | Ad Spend

This interactive visualization is also useful for explaining why correlation changes as the pattern of points changes:

Drag points

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Linear correlation is positive ($r = 0.98$).

NegativeNear zeroPositive

Give feedback

Emphasize:

Correlation does not prove causation.

Advertising and sales being correlated doesn't by itself prove that advertising caused the sales increase.

5. Regression

Now turn the correlation question into a prediction question:

"If we spend ₹50,000 on advertising, what sales might we expect?"

First create:

Insert → Scatter Plot

X-axis = Advertising Spend

Y-axis = Sales

Add:

Trendline → Linear → Display Equation + R^2

You might get something like:

$$\text{Sales} = 14.2 \times \text{Advertising} + 180,000$$

Explain:

- **Intercept** = predicted sales when advertising is zero
- **Slope** = estimated change in sales for each ₹1 change in advertising
- **R²** = proportion of observed variation in Sales explained by the fitted model, within the model assumptions

Then:

Data → Data Analysis → Regression

Set:

Y Range = Sales

X Range = Advertising Spend

Teach students to identify **Coefficients, R Square, p-values and residuals**.

$$\hat{y} = b_0 + b_1x$$

$$\hat{y} = 8.49 - 0.54x$$

$R^2 = 0.72$ · b_0 = intercept · b_1 = slope · least squares minimizes squared vertical residual gaps.

Focus

DataLinePredictionsResiduals

Generate new data

Drag a point to refit the line.

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Give feedback

Then extend to multiple regression:

Sales = f(Advertising, Price, Units)

This is a nice bridge from Excel analytics into machine learning.

6. Moving Average

Now move to time-series analysis.

Business question:

"Daily sales fluctuate heavily. How can management see the underlying trend?"

For a 3-day moving average:

=AVERAGE(F2:F4)

Then drag down:

Day 1 400,000

Day 2 600,000

Day 3 500,000 → MA = 500,000

Day 4 700,000 → MA = 600,000

Day 5 900,000 → MA = 700,000

Also demonstrate:

Data → Data Analysis → Moving Average

Then plot:

Actual Sales vs Moving Average

Explain that moving averages **smooth short-term fluctuations and make trends easier to see**, though they can lag sudden changes.

Move from Analysis to Decision Making

7. What-If Analysis

Create a simple profit model:

Units Sold 1,000

Selling Price ₹500

Variable Cost ₹300

Fixed Cost ₹100,000

Revenue:

=Units*Price

Profit:

=(Units*(Price-VariableCost))-FixedCost

Current profit:

$1000 \times (500 - 300) - 100000 = ₹100,000$

Now ask:

"What happens to profit if price changes?"

Use:

Data → What-If Analysis → Data Table

Put different prices:

₹400

₹450

₹500

₹550

₹600

₹650

Excel calculates profit for every price.

Then teach a **two-variable Data Table**:

Selling Price × Units Sold → Profit

This is excellent for sensitivity analysis.

8. Goal Seek

Reverse the previous problem.

Instead of:

"If I sell 1,000 units, what is my profit?"

Ask:

"How many units must I sell to earn ₹500,000 profit?"

Go to:

Data → What-If Analysis → Goal Seek

Set cell: Profit

To value: 500000

By changing: Units Sold

This usually creates an immediate "aha" moment because Excel works **backwards from the business target**.

Other examples:

What selling price gives 20% margin?

What monthly sales are required to break even?

What interest rate makes EMI ₹50,000?

9. Scenario Manager

Now introduce uncertainty.

Create three scenarios:

Variable	Best Case	Expected	Worst Case
Units	1,500	1,000	700
Price	₹550	₹500	₹450
Variable Cost	₹280	₹300	₹340

Go to:

Data → What-If Analysis → Scenario Manager → Add

Create:

Best Case

Expected Case

Worst Case

Changing cells:

Units + Price + Variable Cost

Result cell:

Profit

Then generate **Scenario Summary**.

Explain the difference:

Goal Seek: "What input gets me this result?"

Scenario Manager: "What happens under different combinations of assumptions?"

Data Table: "How sensitive is my result to one or two variables?"

Data Preparation & BI

10. Loading Data

Now deliberately give students data outside Excel.

Have three files:

January_Sales.csv

February_Sales.csv

March_Sales.csv

Teach:

Data → Get Data → From File → From Text/CSV

Then:

Load

versus

Transform Data

Explain that **Load** brings data into Excel, whereas **Transform** opens Power Query so it can be cleaned before loading.

Also demonstrate importing from:

CSV / TXT → Excel Workbook → Folder → Web → Database

11. Power Query Transformations

This should be one of the most hands-on parts.

Give intentionally dirty data:

" NORTH "

"north"

"North"

"HYDERABAD "

₹12,000

NULL

15/09/26

In **Power Query Editor**, teach:

Remove Columns → Remove Rows → Remove Duplicates → Replace Values → Trim → Clean → Change Data Type → Split Column → Merge Columns → Fill Down → Group By → Pivot → Unpivot → Merge Queries → Append Queries

Example:

Raw:

Customer_Name = " ABC RETAIL "

Apply:

Transform → Format → Trim

Result:

ABC RETAIL

Then transform:

north

NORTH

North

to:

North

North

North

The big lesson:

Don't manually clean the same monthly report every month. Build the transformation once and refresh it.

12. PivotTable

Once clean data is loaded:

Insert → PivotTable

Start with a management question:

"Show sales and profit by region and product."

Configure:

Rows → Region

Columns → Product

Values → Sum of Sales

Values → Sum of Profit

Filters → Year

Then teach:

Sort → Filter → Value Field Settings → % of Grand Total → Group Dates → Slicers → Timeline → PivotChart

A particularly useful exercise:

"Build an interactive regional sales dashboard without writing a single formula."

Students create:

PivotTables + PivotCharts + Slicers

13. Power Query + PivotTable Together

This is where the pieces connect.

Show this architecture:

CSV Files → Power Query → Clean/Transform → Data Model/Excel → PivotTable → Dashboard

Then add a new month's CSV.

Instead of rebuilding everything:

Data → Refresh All

The dashboard updates.

This demonstrates a basic automated analytics pipeline inside Excel.

14. Live Data Connection and Analysis

End with something more advanced.

Explain the difference between:

Import/Load: Excel stores a copy/snapshot of the data.

Connected data: Excel maintains a connection to an external data source that can be refreshed.

Demonstrate a source such as:

Data → Get Data → From Database → SQL Server

or a suitable web/API source.

Conceptually:

Source System → Power Query/Connection → Excel Data Model → PivotTable/Dashboard

Then show:

Data → Queries & Connections → Properties → Refresh

Depending on the connector and Excel version, refresh behavior can vary, so I would describe this as **refreshable/connected analysis** rather than implying that every connection is continuously real-time.